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Ing. Michal Štěpán

Senior GenAI Engineer · LLM applications, RAG, evaluation, MCP · Azure & GCP

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📍 Sviadnov (Ostrava), Czechia · remote / hybrid

🌐 mxstex.github.io/MyWebPage

🗨 Czech native · English B2–C1 (Cambridge FCE)

Profile

GenAI engineer who builds LLM applications end to end: RAG on Azure AI Search since 2025, an LLM working alongside deterministic rules, evaluation on golden datasets, MCP servers for AI coding assistants, and delivery to Azure and GCP. In my own time I build the same way as Tojin Games, together with Claude Code. Before ABB: freelance data and backend engineering, energy analytics at Siemens and ABB, and eight years at a district-heating utility, the last ones leading more than fifteen people. Nuclear engineer by training. I want to keep building LLM applications and agents, with a larger share of the design and technical lead.

RAG since 2025

Azure AI Search · LangGraph · Document Intelligence

Evaluation first

golden datasets, acceptance and stress tests

MCP servers

tools for Claude Code and GitHub Copilot

Tojin Games

8 games, 5 live on Cloud Run, 40+ videos

Core skills

● GenAI & LLM applications

Azure OpenAI / AI Foundry · RAG on Azure AI Search
vector + hybrid retrieval
LangChain · LangGraph (mostly 2025) · ChromaDB
Document Intelligence · structured outputs
LLM + deterministic rules · golden datasets & evaluation
MCP servers · LLM agents on Ollama · pgvector memory
Claude Code · GitHub Copilot

● Cloud & delivery

Azure Container Apps · Container Registry
Key Vault · managed identity · Log Analytics
Blob & Table Storage · Azure Pipelines (dev → prod)
GCP Cloud Run · Secret Manager · Terraform (GCP) · Docker
OAuth 2.0 / JWT

● Programming & testing

Python · FastAPI · Pandas · NumPy / SciPy · NiceGUI
Pytest · Playwright · load & stress tests · SQL / PostgreSQL
JavaScript · Canvas · WebGL2 · Kotlin + Jetpack Compose

● Worked with earlier

Dagster · Airflow · DuckDB · BigQuery · RabbitMQ
Kubernetes · GitLab CI · GitHub Actions · Flask · Streamlit
AWS S3 · Lambda · Azure Functions · Grafana · Power BI
ARIMA / SARIMA

Professional experience

Senior GenAI Engineer · ABB s.r.o.

Sep 2025 – present

- RAG since 2025: retrieval over Azure AI Search with vector embeddings and hybrid retrieval, LangChain and LangGraph workflows, ChromaDB, and Azure Document Intelligence to turn documents into indexable content. In 2026 wrote the architecture documentation for a RAG assistant (Azure AI Search, LangGraph, Azure OpenAI).
- AI-assisted validation of engineering spreadsheets: built the evaluation (golden datasets, acceptance evaluation, performance and stress tests) that every change is measured against.
- Backend in FastAPI: a YAML rule engine that runs deterministic checks next to the LLM validators, switchable model profiles (Azure OpenAI or a local model through Ollama) with structured outputs, quotas and session cleanup; benchmarked local against cloud models.
- Delivery: Azure Pipelines for dev, test, stage and prod; Docker images in Container Registry, Azure Container Apps, Key Vault secrets through managed identity, Log Analytics.
- Built a proof-of-concept MCP server (Python, FastMCP) that gives AI coding assistants (Claude Code, GitHub Copilot) access to Azure DevOps work items, with a mock API and tests.

STACK Python · FastAPI · Azure OpenAI / AI Foundry · Azure AI Search · LangChain · LangGraph · Document Intelligence · Pandas · Pytest · Docker · Azure Container Apps · Key Vault · Azure Pipelines · MCP · Ollama

Freelance Software / Data Engineer

Sep 2022 – Sep 2025

DTSE, Landis+Gyr, Vendavo and other clients in energy, telecom and manufacturing · remote, international teams

- Designed and built end-to-end data workflows in Python, orchestrated with Dagster and Airflow, stored in PostgreSQL, DuckDB and BigQuery.
- Built backend components for high-volume ingestion and message-based processing with RabbitMQ, and REST services.
- Migrated services from Kubernetes to cloud-native GCP (Cloud Run, GCS, BigQuery).
- Maintained CI/CD in GitLab CI and GitHub Actions with unit, integration and end-to-end tests in Pytest.
- Delivered high-performance processing pipelines for recommender systems (multiprocessing, optimisation); L2 support for production systems.

STACK Python · Dagster · Airflow · PostgreSQL · DuckDB · BigQuery · RabbitMQ · GCP · Kubernetes · GitLab CI · GitHub Actions

Senior Software Developer · Siemens s.r.o.

Jan 2021 – Aug 2022

- Developed an Energy Management System consolidating and analysing data from many sources.
- Implemented ARIMA and SARIMA models for time-series forecasting.
- Mentored junior developers.

STACK Python · Flask · FastAPI · Docker · PostgreSQL · Grafana · Power BI · RabbitMQ · AWS S3 · AWS Lambda

Software Project Engineer · ABB s.r.o.

Oct 2018 – Jan 2021

- Developed a water and energy consumption evaluation system; designed and implemented predictive-analytics features.

STACK Python · Docker · PostgreSQL · Azure Functions · Azure Blob Storage · Grafana · Power BI

IT Specialist · Power Engineer · Technical Deputy · DISTEP a.s.

May 2010 – Oct 2018

- As technical deputy, led an operations team of more than fifteen people in water and heat distribution.
- Improved resource efficiency through optimised calculations and monitoring; streamlined data collection with PDA barcode readers (MS SQL, VBA).

Tojin Games · my own studio, in my own time

Independent science, space and simulation games · YouTube youtube.com/@tojin111 · live builds on the portfolio site

- **Games:** eight science, space and strategy games built solo, plus an early online prototype of Jednota; five are live in the browser on Google Cloud Run. Hand-written engines for Newtonian gravity, Kepler orbits with WebGL2 rendering, plasma confinement (Biot-Savart) and the Schrödinger equation (split-step Fourier); in Gravity, tests keep the JavaScript engine bit-identical to the Python reference and the Kotlin one within 1e-9.
- **Jednota**, a 4X strategy game run by LLM agents (~73,000 lines of Python, about 1,300 tests): strategist, envoy and fleet-commander agents on local LLMs through Ollama, with JSON-schema output and no agent framework; deterministic rules validate every model decision (accepted 93/93 strategic, 80/88 diplomatic, 79/86 tactical in a measured five-empire campaign); memory across games through pgvector RAG.
- **Platform:** an OAuth 2.0 to RS256 JWT entitlement service (FastAPI, Cloud Run, Secret Manager, Terraform, 54 tests) gives five games one shared login and access tiers; every service scales to zero under a one-instance cost cap.
- **Video pipeline:** code-driven 4K capture of real game sessions (Playwright, Chrome DevTools Protocol, ffmpeg/NVENC), captions from on-screen events, local TTS, loudness-checked masters and automated validation, plus local ComfyUI models (FLUX, Wan 2.2): more than forty videos published in September 2026.
- **How:** developed with Claude Code, which co-wrote about 70% of the commits; the results are checked by test suites, level verifiers and bots that play the games through.

Education & certificates

Ing. (MSc), Electrical Engineering and Computer Science · 2012

VŠB – Technical University of Ostrava. Thesis: VBA Utilization for Power Engineering Data Processing and Presentation

Bc., Nuclear Engineering · 2009

Czech Technical University in Prague, Faculty of Nuclear Sciences and Physical Engineering. Thesis: Physical Analysis of Fast Reactor Cores in Up-to-Date Tendencies

Research project · 2006

Institute of Plasma Physics, Czech Academy of Sciences: electrostatic turbulence in the edge plasma of the CASTOR tokamak

Microsoft AZ-305 · AZ-400

Azure infrastructure design and DevOps solutions · training

Cambridge English: First (FCE)

Grade B · CEFR B2 · 2019